

ENTHUSIAST ADVANCE COURSE

PHASE : MEC-I (B)

TARGET : PRE MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 16-07-2023

ANSWER KEY

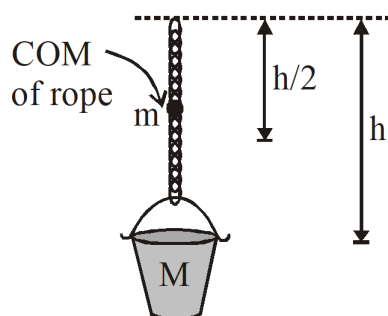
Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	1	3	1	1	2	3	4	1	1	1	1	4	2	2	1	3	2	1	2	2	1	2	1	3	2	3	1	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	2	2	2	3	2	2	1	2	3	3	2	4	2	2	4	1	3	3	3	1	3	1	2	2	3	2	2	4	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	3	1	2	2	2	4	3	1	3	3	4	2	3	2	2	3	3	2	1	3	2	4	1	3	3	3	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	3	3	4	3	3	2	3	4	2	2	1	1	3	1	3	3	1	3	2	1	2	3	3	3	2	1	4	1	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	4	3	2	2	2	3	2	4	1	1	3	3	4	1	3	2	3	2	3	4	1	4	3	3	4	1	4	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	4	1	1	1	4	1	3	2	4	4	1	3	4	4	1	2	4	1	1	2	1	3	4	3	4	4	3	1	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	4	1	3	3	2	4	2	3	3	1	3	2	1	3	4	4	4	1	4										

HINT - SHEET

SUBJECT : PHYSICS

SECTION - A

1. Ans (2)



$$w = mg \frac{h}{2} + Mgh$$

$$w = \left(M + \frac{m}{2} \right) gh$$

3. Ans (1)

$$P = Fv = mav = m \frac{dv}{dt} v$$

$$\int_0^v v dv = \frac{P}{m} \int_0^t dt$$

$$\frac{v^2}{2} = \frac{P}{m} t$$

$$v = \sqrt{\frac{2Pt}{m}}$$

4. Ans (3)

72% of 50 Joule

$$\frac{50}{100} \times 72 = 36$$

$$36 = \frac{1}{2} mv^2$$

$$V = \sqrt{72} = 8.5 \text{ m/sec}$$

5. Ans (1)

$$F = 50 \left(\frac{3\hat{i} + 4\hat{j}}{5} \right)$$

$$F = 30\hat{i} + 40\hat{j}$$

$$\vec{s} = 2\hat{i} - \hat{j} - \hat{k}$$

$$W = \vec{F} \cdot \vec{s} = 60 - 40 = 20 \text{ J}$$

6. Ans (1)

$$T.E = mgh$$

$$\text{its height at } h/6 = \frac{5h}{6}$$

$$PE = mg \frac{5h}{6} = \frac{5}{6} mgh$$

$$KE = mgh - \frac{5}{6} mgh$$

$$= \frac{mgh}{6}$$

$$KE : PE \Rightarrow 1 : 5$$

7. Ans (2)

$$B' = n^2 B \Rightarrow B' = (4)^2 B = B' = 16B$$

8. Ans (3)

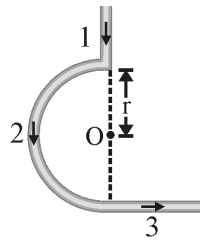
Magnetic field due to different parts are

$$B = 0$$

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{r} \odot$$

$$B_3 = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} \odot$$

$$B_{\text{net}} = B_2 + B_3 = \frac{\mu_0 i}{4r} + \frac{\mu_0 i}{4\pi r}$$



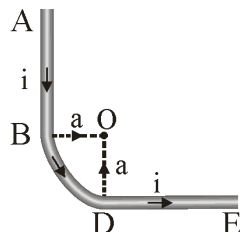
9. Ans (4)

The field at O due to AB is $\frac{\mu_0}{4\pi} \cdot \frac{i}{a} \hat{k}$ and that

due to DE is also $\frac{\mu_0}{4\pi} \cdot \frac{i}{a} \hat{k}$

However the field due to BCD is $\frac{\mu_0}{4\pi} \cdot \frac{i}{a} \left(\frac{\pi}{2} \right) \hat{k}$

Thus the total field at O is $\frac{\mu_0}{4\pi} \cdot \frac{i}{a} \left(2 + \frac{\pi}{2} \right) \hat{k}$

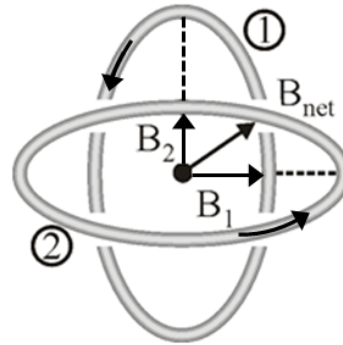


10. Ans (1)

$$B_1 = B_2 = B = \frac{\mu_0}{4\pi} \times \frac{2\pi i}{r}$$

$$B_{\text{net}} = \sqrt{2}B$$

$$\Rightarrow \frac{B}{B_{\text{net}}} = \frac{1}{\sqrt{2}}$$



14. Ans (4)

$$\delta = 180 - 2i$$

$$120 = 180 - 2\theta$$

$$\theta = 30^\circ$$

17. Ans (1)

$$1 \times \sin 60^\circ = \mu_{\text{oil}} \times \sin 30^\circ$$

$$\mu_{\text{oil}} = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$$

19. Ans (2)

$$\theta = (\mu_b - \mu_R)A$$

$$= (1.659 - 1.641) \times 5$$

$$= 0.09^\circ$$

21. Ans (2)

$$P_1 = \frac{\frac{3}{2} - 1}{R} = \frac{1}{2R}$$

$$P_2 = \frac{1 - \frac{3}{2}}{-R} = \frac{1}{2R}$$

Both powers are positive

it means converging

22. Ans (2)

$$\frac{1}{10} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{7.5} - \frac{1}{-R} \right)$$

$$\Rightarrow \frac{1}{10} = \left(\frac{1}{2} \right) \left(\frac{1}{7.5} + \frac{1}{R} \right)$$

$$\frac{1}{R} = \frac{1}{5} - \frac{1}{7.5} = \frac{1}{15}$$

$$R = 15 \text{ cm}$$

23. Ans (1)

$$\text{Angular magnification } |m| = \frac{f_0}{f_e} = 5 \Rightarrow f_0 = 5f_e$$

$$\text{distance between objective and eye-piece} = f_0 + f_e$$

$$f_0 + f_e = 36$$

$$5f_e + f_e = 36 \Rightarrow f_e = 6 \text{ cm}$$

$$\text{and } f_0 = 30 \text{ cm}$$

24. Ans (2)

$$\tan(i_p) = \mu$$

$$\tan(i_p) = \sqrt{3}$$

$$i_p = 60^\circ$$

26. Ans (3)

$$\lambda \propto \frac{1}{\sqrt{T}} \Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}}$$

$$\Rightarrow \frac{\lambda}{\lambda_2} = \sqrt{\frac{1200}{300}}$$

$$\lambda_2 = \frac{\lambda}{2}$$

27. Ans (2)

$$E = \phi_0 + KE_{\max} \quad KE_{\max} = 0.9 \text{ eV}$$

$$E = 3.1 + 0.9 \quad \phi_0 = \frac{12400}{4000 (A^\circ)}$$

$$E = 4 \text{ eV} \quad \phi_0 = 3.1 \text{ eV}$$

$$E = \frac{12400}{\lambda}$$

$$\lambda = \frac{12400}{E} = 3100 \text{ \AA} \text{ or } 310 \text{ nm}$$

28. Ans (3)

$$E_{\text{photon}} = \left(-\frac{13.6}{9} \right) \text{ eV} - (-13.6 \text{ eV}) \approx 12.1 \text{ eV}$$

$$eV_0 = 12.1 \text{ eV} - 5.1 \text{ eV} \Rightarrow V_0 = 7 \text{ V}$$

31. Ans (4)

$$\text{As } 2\pi r = n\lambda = 2 \times 10^{-6} \text{ m}$$

32. Ans (2)

$$\lambda_1 = 1 \text{ \AA}$$

$$\lambda_2 = 0.5 \text{ \AA}$$

$$\therefore V = \frac{150}{\lambda_e^2}$$

$$V_1 = \frac{150}{(1)^2} = -150 \text{ volt}, E_1 = 150 \text{ eV}$$

$$V_2 = \frac{150}{(0.5)^2} = \frac{150}{0.25} = 600 \text{ volt}, E_2 = 600 \text{ eV}$$

$$\Delta E = E_2 - E_1 = 600 - 150 = 450 \text{ eV}$$

33. Ans (2)

TG: @Chalnaayaaar

$$\Delta m = 0.007 \text{ gm}$$

$$Q = \Delta m \times c^2 = 0.007 \times 10^{-3} \times 9 \times 10^{16}$$

$$Q = 63 \times 10^{10} \text{ J}$$

SECTION - B

36. Ans (2)

By COME law

$$mE_i = mE_f$$

$$-\frac{m}{3}g\frac{L}{6} + 0 = -mg\frac{L}{2} + \frac{1}{2}mv^2$$

$$v^2 = \frac{8Lg}{9} \Rightarrow v = \frac{2}{3}\sqrt{2gL}$$

38. Ans (1)

$$K = \frac{1}{2} \frac{mv^2}{f}$$

$$= \frac{1}{2} \frac{(v\rho)v^2}{f}$$

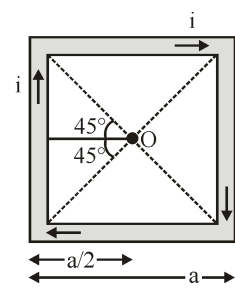
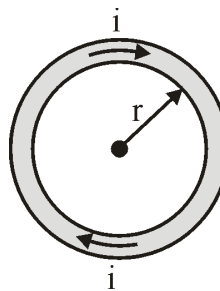
$$= \frac{1}{2} \frac{(A\ell)\rho v^2}{f}$$

$$= \frac{1}{2} \rho A v^3$$

39. Ans (2)

(b) Circular coil

Square coil



$$\text{Length } L = 2\pi r \quad \text{Length } L = 4a \Rightarrow a = \frac{L}{4}$$

Magnetic field at the centre of circular coil

$$B_{\text{circular}} = \frac{\mu_0}{2} \cdot \frac{i}{r} = \frac{\mu_0}{2} \frac{2\pi i}{L}$$

Magnetic field at the centre of square coil

$$B_{\text{square}} = \frac{4\mu_0 i}{\pi \left(\frac{L}{4}\right)} \sin \frac{\pi}{4} \tan \frac{\pi}{4} = \frac{8\sqrt{2}\mu_0 i}{\pi L}$$

$$\text{Hence, } \frac{B_{\text{circular}}}{B_{\text{square}}} = \frac{\pi^2}{8\sqrt{2}}$$

40. **Ans (3)**

Magnetic field at any lying on the current carrying straight conductor is zero.

Here H_1 = Magnetic field at M due to current in PQ.

H_2 = Magnetic field at M due to QR
 + magnetic field at M due to QS
 + magnetic field at M due to PQ

$$H_2 = 0 + \frac{H_1}{2} + H_1 = \frac{3}{2}H_1 \Rightarrow \frac{H_1}{H_2} = \frac{2}{3}$$

43. **Ans (4)**

Distance $|V_0| + |u_e|$ use lens formula for objective

$$V_0 = 6 \text{ cm}$$

lens formula for eye piece $|u_e| = 5$
 distance = $6 + 5 = 11 \text{ cm}$

45. **Ans (2)**

$$\text{Energy of one photon} = \frac{hc}{\lambda} = 4 \times 10^{-19} = E$$

$$(n) (E) = \text{Energy} = 10^{-7} \text{ J}$$

$$(n) (4 \times 10^{-19}) = 10^{-7} \text{ J}$$

$$n = 2.5 \times 10^{11}$$

46. **Ans (4)**

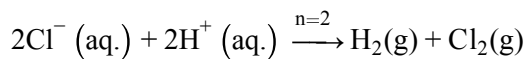
$$F = \frac{P}{C}$$

$$= \frac{IA}{C} = \frac{1400 \times 1.5}{3 \times 10^8} = 7 \times 10^{-6} \text{ N}$$

SUBJECT : CHEMISTRY

SECTION-A

56. **Ans (3)**



$$10^{-2} \text{ M} \quad 10^{-1} \text{ M} \quad 1 \text{ atm} \quad 1 \text{ atm}$$

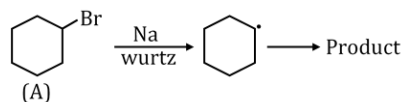
$$E = -1.36 - \frac{0.06}{2} \log \frac{1 \times 1}{10^{-4} \times 10^{-2}}$$

$$E = -1.5 \text{ V}$$

62. **Ans (3)**

Reaction proceeds through S_N^2 mechanism.

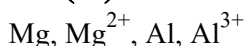
64. **Ans (1)**



74. **Ans (2)**

Left to right size decreases.

75. **Ans (3)**



$\text{Mg} > \text{Mg}^{2+}, \text{Al} > \text{Al}^{3+}$ Neutral > cation of same element (size)

Largest $\text{Mg} > \text{Al}$

$\text{Mg}^{+2} > \text{Al}^{+3}$ smallest

76. **Ans (2)**

Correct order $\text{S} > \text{Se} > \text{Te} > \text{O}$

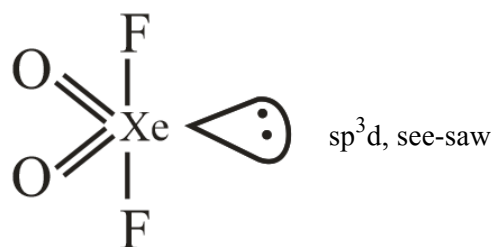
77. **Ans (2)**

Electron gain enthalpy order is $\text{Cl} > \text{F} > \text{Br}$

78. **Ans (3)**

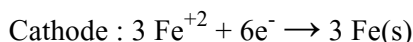
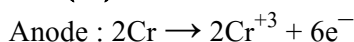
Electron deficient and molecule showing expanded octet cannot follow octet rule.

79. **Ans (3)**

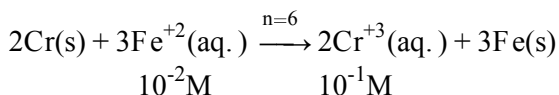


SECTION-B

90. **Ans (3)**



Net cell reaction



$$E_{\text{cell}} = (-0.42 + 0.72) - \frac{0.06}{6} \log \frac{10^{-2}}{10^{-6}}$$

$$= 0.3 - 0.01 \log 10^4$$

$$= 0.3 - 0.04 = 0.26 \text{ V}$$

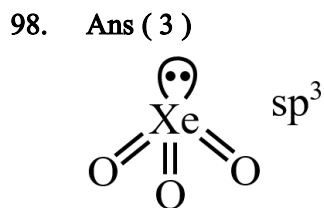
96. **Ans (3)**

$$\text{I.P} \propto \frac{1}{\text{metallic character}} \propto \text{non metallic character}$$

$$\text{I.P} = \text{F} > \text{N} > \text{C} > \text{B} > \text{Si}$$

Non-metallic = $\text{F} > \text{N} > \text{C} > \text{B} > \text{Si}$
 character

97. **Ans (2)**
Acidic Nature $\propto$ O.S. of Non-Metal



SUBJECT : BIOLOGY-I

SECTION-A

101. **Ans (2)**
NCERT XII Pg. # 224
102. **Ans (1)**
NCERT-XII Page No. 255
103. **Ans (1)**
NCERT XII Pg. # 245
104. **Ans (3)**
NCERT XII Pg. # 249
105. **Ans (1)**
NCERT XII Pg. # 243
106. **Ans (3)**
NCERT (XIIth) Pg. # 77
111. **Ans (1)**
NCERT Supplements
115. **Ans (3)**
NCERT-XII, Pg. # 87
118. **Ans (4)**
NCERT (XII) Pg. # 88
119. **Ans (1)**
NCERT Pg#84
120. **Ans (1)**
NCERT Pg. # 85
121. **Ans (3)**
NCERT Pg # 89

122. **Ans (4)**
NCERT-XII (E), Pg#89,90
NCERT-XII (H), Pg#98,99
123. **Ans (3)**
NCERT XIIth Pg # 235
126. **Ans (2)**
NCERT Pg. # 235, Para 13.2.4

SECTION-B

136. **Ans (3)**
NCERT (XII) Pg. # 223
143. **Ans (4)**
NCERT Pg. # 234
146. **Ans (4)**
NCERT XII, Page No. # 227, 229

SUBJECT : BIOLOGY-II

SECTION-A

166. **Ans (1)**
NCERT-XII, Pg. # 262
167. **Ans (2)**
NCERT Pg. # 257 16.1.1
169. **Ans (1)**
NCERT Pg. # 286
170. **Ans (1)**
NCERT Pg-278
171. **Ans (2)**
NCERT Pg#282

SECTION-B

190. **Ans (3)**
NCERT-XI, Pg # 264, 1st para